1. A company suspects information leakage. Examine system files, browser data, and metadata to identify digital footprints using Metadata2Go.
2. A company suspects information leakage. Investigate system files, browser data, and storage artifacts to identify digital evidence using FTK Imager.
3. A company suspects information leakage. Analyze system files, browser artifacts, and metadata to identify digital footprints using Autopsy.
4. An organization wants to evaluate its system security. Perform vulnerability assessment and identify security weaknesses using Nessus / OpenVAS.
5. A system shows signs of unauthorized access. Demonstrate password cracking techniques through active online attacks and assess associated security risks using John the Ripper.
6. A system shows signs of unauthorized access. Demonstrate brute-force password attacks to evaluate weak authentication mechanisms using Hydra.
7. An attacker gains user-level access. Analyze privilege escalation techniques using exploitation frameworks and recommend suitable countermeasures with Metasploit Framework.
8. Confidential information must be hidden securely. Apply steganography techniques to conceal data within media files using Steghide.
9. Employees are targeted through deceptive messages. Analyze social engineering attacks and phishing techniques using Social-Engineer Toolkit.
10. A network administrator tests intrusion detection systems. Analyze IDS evasion techniques through stealth scanning using Nmap.

11. A user session behaves abnormally. Apply session hijacking techniques and examine their impact on authentication and system security using Wireshark / Ettercap.
12. A web service becomes unavailable during heavy traffic conditions. Analyze the impact of DoS and DDoS attacks on system performance and availability using Wireshark.
13. Smart devices show suspicious activity. Analyze IoT devices using reconnaissance techniques to identify security exposures using Shodan / Nmap.
14. Secure data transmission is required. Apply encryption techniques to protect files and text messages using OpenSSL / GnuPG.
15. Encrypted evidence is obtained during investigation. Analyze encrypted data using cryptanalysis tools to identify weaknesses using Hashcat / CrypTool.